

Verizon EDA - Yining Mao

Business Problem and Objective: Verizon Communications Inc., the second-largest U.S. wireless carrier and a Fortune 100 company, generates significant revenue by financing customer devices through monthly payment plans. If customers fail to make their payments, Verizon suffers losses from the unpaid device balance as well as the time and resources spent servicing the loan. The goal is to maximize profit for Verizon by building a machine learning model that identifies customers with a higher chance of defaulting on their loans by analyzing historical customer data.

Using a logistic regression model: From a business standpoint, this model helps Verizon approve reliable customers who are likely to pay on time while flagging higher-risk applicants for additional review. It also allows the company to adjust approval thresholds to find the right balance between increasing sales and minimizing potential losses, ultimately maximizing overall profit. By looking at patterns in historical data, the model estimates the probability that a customer will become default on their payments. The model examines the relationship between the default variables and other factors. Logistic regression is a good fit because it predicts outcomes that have two possibilities, either default or no default, and provides clear insights into what factors will increase risk.

From a technical perspective, the model uses logistic regression to estimate the probability that a Verizon customer will default on their device financing plan. Logistic regression handles binary outcomes well (default = 1, non-default = 0) and provides interpretable coefficients to show how each factor influences the likelihood of default. The dependent or target variable is **del90** (90+ days delinquent), which has a binary outcome. The predictors include **FICO** (Customer FICO credit score), **device_cost** (Price of the device being financed), **down_payment** (initial payment made by customer), **payment_type** (method for the downpayment), and **borrower_age** (customer age in years).

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1(fico) + \beta_2(deviceCost) + \beta_3(downPayment) + \beta_4(borrowerAge) + \beta_{cash}(cash) + \beta_{credit}(credit)$$

Where

- p_i : Probability that customer i defaults
- β_0 : Intercept
- β_1 β_2 β_3 and β_4 : Coefficients for numeric features
- β_{cash} and β_{credit} : Coefficients for the payment type as categorical features

Model Implementation: Before running the model, the categorical variable payment_type is converted into a binary variable through one-hot encoding, where each category is represented by a 1 if it applies, and 0 otherwise. Bank Transfer is dropped from the model to avoid multicollinearity, as it is a neutral benchmark between cash (risky) and credit (safer). The data is then split into 70% training and 30% testing. The logistic regression model is fitted with maximum likelihood estimation (LogisticRegression() in Python from scikit-learn). After training, the model outputs a probability score between 0 and 1 for each applicant, representing their likelihood of default. The random state is

set at 42 to guarantee same reproducible result and the max iteration is set at 2000 so there's enough iterations. The decision threshold can be adjusted by Verizon, for example, at 0.50, the model is stricter and approves more customers, while at 0.30 it flags more potential defaulters, allowing Verizon to choose the cutoff that best balances sales and credit risk. The model's performance is then evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and ROC-AUC.

Result and Interpretation:¹ The model coefficient values show how each variable influences the log-odds of default while controlling for other factors. A positive coefficient increases the likelihood of default, while a negative coefficient decreases it. Customers who pay by cash have a slightly higher risk of default (+0.096), while those who pay by credit card have a lower risk (−0.173). FICO score has the strongest negative effect (−0.266), meaning higher credit scores significantly reduce the odds of default. In summary, customers with lower FICO scores, smaller down payments, and cash payment methods are more likely to default, while those with higher credit scores, larger down payments, or who pay with credit cards tend to be more reliable.

The model performance is assessed using the following metrics that can be easily used to compare with other models. It demonstrates two decision thresholds: 0.5 (default) and 0.3 (more cautious)

Metric	Threshold 0.50	Threshold 0.30	What It Means for Verizon
Accuracy	0.75 (75%)	0.68 (68%)	At 0.50, the model correctly predicts about 3 out of 4 cases overall. Accuracy drops slightly at 0.30 because the model becomes more cautious and flags more customers as risky.
ROC-AUC	0.60	0.60	The model can correctly rank a random defaulter above a random non-defaulter 60% of the time, which is 10% better than random guessing
Precision	1.00 (100%)	0.33 (33%)	At 0.50, every customer flagged as risky actually defaulted, but the model rarely flags anyone. At 0.30, it identifies more defaulters but also includes some false alarms, which is acceptable in business to reduce losses.
Recall	0.001 (0.1%)	0.27 (27%)	At 0.50, the model misses almost all true defaulters. Lowering to 0.30 greatly improves the model's ability to catch risky customers.
F1-Score	0.002	0.30 (30%)	The balance between precision and recall improves sharply at 0.30, meaning the model becomes more useful for identifying high-risk borrowers at a lower threshold

Using Model Results to measure Verizon's Business Benefit to maximize profit

By setting a decision cutoff (for example, 0.30), Verizon can decide which customers to approve or decline. Comparing the model's predictions with actual repayment outcomes creates a confusion matrix with four possible results:

- Risky customers correctly identified and declined: losses avoided.
- Safe customers correctly approved: profit earned.
- Safe customers wrongly rejected: missed profit.

¹ The detailed regression results are in the appendix

- Risky customers wrongly approved: losses incurred.

The total business value of the model can be estimated using:

$$\Delta \text{Value} = (\text{Safe Correctly Approved} \times \text{Profit}) + (\text{Risky Avoided} \times \text{Loss}) \\ - (\text{Risky approved} \times \text{Loss}) - (\text{Safe Wrongly Rejected} \times \text{Profit})$$

By applying the model's predictions to the ΔValue equation, Verizon can see how profit changes at different approval cutoffs. A lower cutoff approves more customers and drives sales but increases losses from risky borrowers, while a higher cutoff reduces defaults but may reject safe, profitable customers. Verizon can find the optimal threshold that maximizes total profit while effectively managing credit risk.

Why this model: Logistic regression was chosen because it offers the right balance of accuracy, speed, and simplicity for businesses to operate. It clearly shows how factors such as FICO score, down payment, and payment method affect the likelihood of default, making it easy for Verizon managers to understand trust and use it to calculate profit. Unlike more complex models, it's fast to train, easier to explain, and works well for binary outcomes like default versus no default. Customers with higher risk scores can be flagged immediately for additional review.

Data limitations, Bias, and Artificial signatures

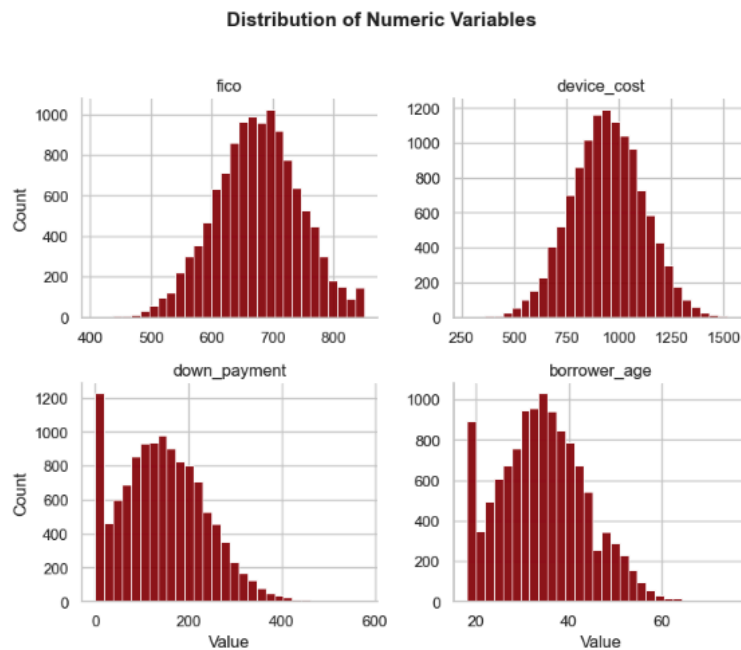


Figure 1 — Distribution of Numeric Variables shows a lack of diversity and over-representation of mid-range borrowers.

FICO scores cluster around the average, with underrepresented high and low credit borrowers, which suggests that the model may struggle to capture the full range of credit risk and could perform less accurately for borrowers outside the average range. The device costs appear to be very evenly distributed and symmetric, implying equal representation of low, mid, and high prices, where in reality, most customers buy mid-range devices due to affordability. Down payments are concentrated near the low end, with a noticeable spike at \$0. This pattern suggests many customers made little or no upfront

payment. Since down payment size is often linked to credit risk and financial stability, the limited variation makes it harder for the model to understand how upfront commitment affects default risk. Borrower age centers heavily around middle-aged customers, with fewer younger or older applicants.

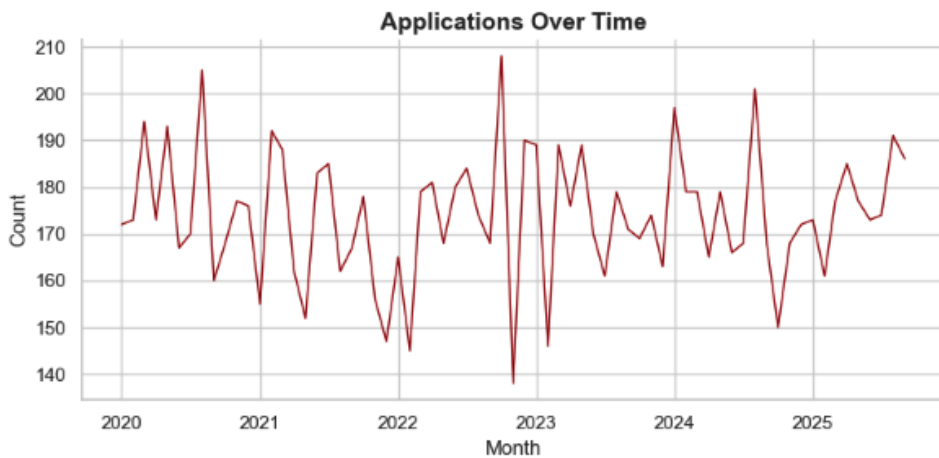


Figure 2 — Applications Over Time further shows a stable number of applications throughout 2020–2025, suggesting a lack of variation that would normally occur with market cycles or tightening approval standards.

Artificial Signature #1 The application volume over time shows a flat, randomly fluctuating pattern between 2020 and 2025, with no visible seasonal spikes, growth trends, or external event effects such as COVID-19 or holiday surges. In real-world data, we would expect higher application activity during product launches and holiday seasons, and lower activity during economic disruptions. The absence of these patterns suggests that the dataset was simulated.

The dataset demonstrates selection bias and imbalance, where it only includes approved financing applications, not those that were initially rejected. This means the model learns patterns based on already-screened customers rather than the full applicant pool. As a result, it may underestimate true default risk, since higher-risk applicants were excluded from the start. Additionally, the response variable, del90, represents whether a customer was ever 90 days delinquent but not necessarily whether they ultimately defaulted. Some loans may still be active or have recently originated, meaning default outcomes are not yet observable, which can also contribute to data bias.

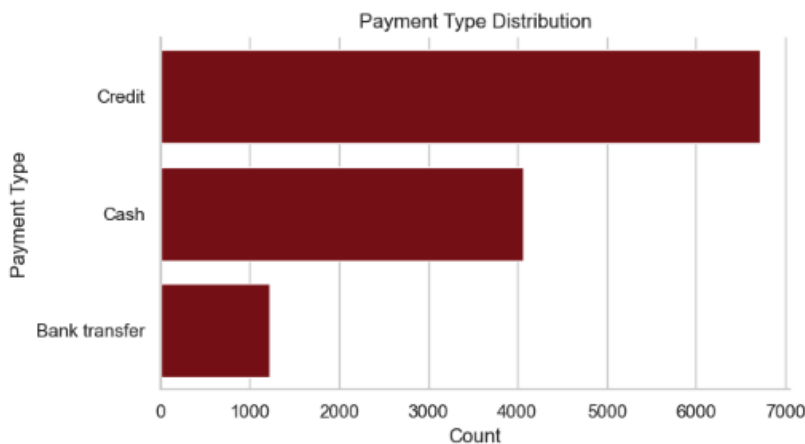


Figure 3 — Credit payments represent the largest share, but the even spacing across payment types and lack of variation indicate possible simulation rather than real customer payment behavior.

The dataset also lacks key financial variables such as interest rate, income, employment type, or total debt, which are critical for modeling affordability and payment burden. Without these, the logistic regression model cannot fully capture why two borrowers with similar credit scores might behave differently. Additionally, the data omit geographic or store-level identifiers, preventing analysis of regional differences in customer demographics or store risk management. In addition, there is no missing value in the dataset, demonstrating unrealistic data quality as missing values are almost always present due to incomplete applications, data entry errors, or a lack of customer information.

Artificial Signature #2 The payment type distribution shows that Credit, Cash, and Bank Transfer have very clean, evenly spaced counts. Such rounded proportions are uncommon in real data. There is also no temporal or regional variation where real payment preferences typically shift over time (e.g., more people enroll in AutoPay in recent years). Verizon plans usually include more options like AutoPay or Direct Debit, so having only three simplified payment types suggests the data was simulated.

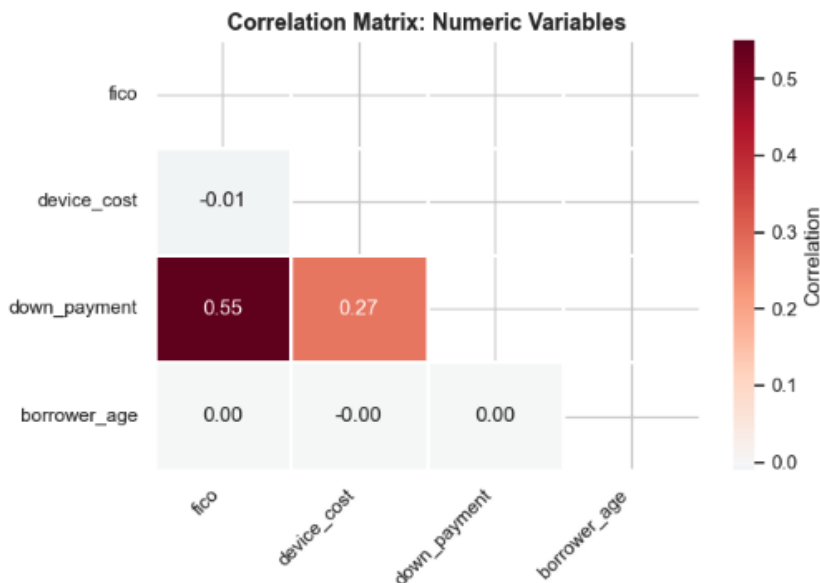


Figure 4 — Correlations show that higher FICO scores align with larger down payments, yet most variables are weakly related, indicating limited behavioral diversity and potential artificial structure in the dataset.

The correlation matrix reveals only a few simple relationships among the numeric variables, with the strongest between down payment and FICO score (0.55) and a moderate link between down payment and device cost (0.27). **Artificial Signature #3** The correlation matrix appears unusually clean and rounded. It's also unrealistic that borrower age shows no relationship with FICO or down payment, since older borrowers usually have higher credit scores and different spending behavior. The overall lack of strong or complex relationships among variables suggests the dataset was generated synthetically rather than collected from real Verizon customers.

Requests for Verizon: To improve measurement accuracy and strengthen the predictive power of the logistic regression model, several additional data elements should be provided by Verizon that can be retrieved within 24 hours.

- Loan Interest Rate and Financing Terms:**

Including each plan's APR, loan length (for example, 12, 24, or 36 months), and payment

schedule information, such as due dates and grace periods, allows the model to better capture how payment pressure and timing affect the likelihood of defaulting. These details help the model better understand how the size and timing of payments influence whether customers keep up with or miss their payments. It increases the accuracy of predicting affordability and also increases the model's ability to explain why customers with the same FICO might have different outcomes.

2. **Customer Relationship History with Verizon:**

Information on whether the borrower is a new or returning customer, the length of their relationship with Verizon, and their previous financing or payment history can provide insight into their loyalty and reliability. Including these details helps the model tell the difference between customers who consistently make their payments and those who are more likely to fall behind. The model will be able to better differentiate between loyal, low-risk customers and the new, untested applicants, thus reducing false positives.

3. **Promotion Participation and Device Information:**

Data on whether a financing plan was associated with a marketing campaign, such as a "\$0 down" offer, and the corresponding device brand or model tier adds important context about customer segments and purchasing patterns. This information improves the model's predictions by helping it recognize how different types of customers tend to behave and manage their payments. Customers attracted by \$0 down promotions might show different repayment habits or risk profiles, and device type can proxy for income level and spending behavior.

The addition of these can help with filling in behavioral, financial, and contextual characteristics of customers, which can increase the model's explanatory power.

Appendix:

Figure A1. Model Coefficients and Feature Effects

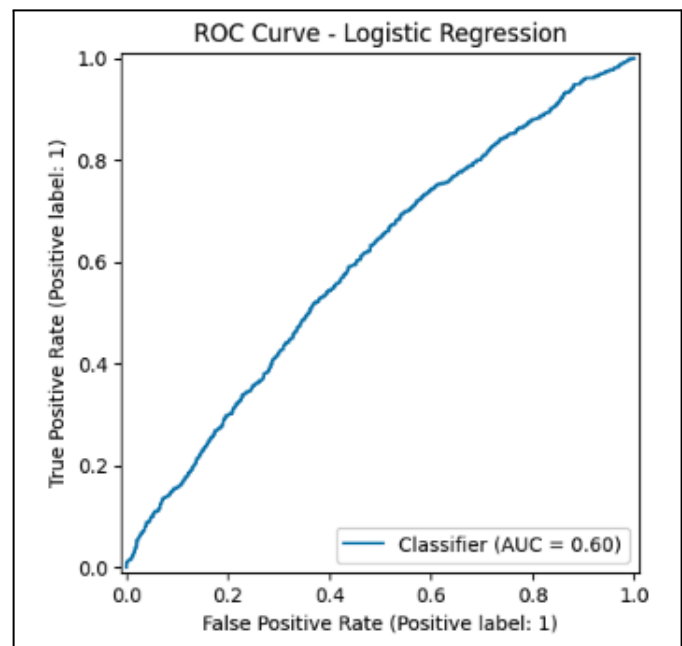
Feature Effects (sorted by log-odds):				
Feature	Coef (log-odds)	Odds Ratio	Direction	
payment_type_Cash	0.096389	1.101188	Increases risk	
device_cost	-0.015419	0.984699	Reduces risk	
down_payment	-0.050829	0.950442	Reduces risk	
borrower_age	-0.100982	0.903950	Reduces risk	
payment_type_Credit	-0.173196	0.840973	Reduces risk	
fico	-0.265943	0.766483	Reduces risk	

Figure A2. Classification Metrics at 0.50 and 0.30 Thresholds

=== Classification @ 0.50 threshold ===				
	precision	recall	f1-score	support
0	0.754	1.000	0.860	2714
1	1.000	0.001	0.002	886
accuracy			0.754	3600
macro avg	0.877	0.501	0.431	3600
weighted avg	0.815	0.754	0.649	3600

=== Classification @ 0.30 threshold ===				
	precision	recall	f1-score	support
0	0.775	0.817	0.796	2714
1	0.328	0.273	0.298	886
accuracy			0.683	3600
macro avg	0.551	0.545	0.547	3600
weighted avg	0.665	0.683	0.673	3600

Figure A3. ROC Curve (AUC = 0.60)



References:

Verizon Logistic Regression Cheat Sheet [Class material]. University of Chicago, MS in Applied Data Science Program.

OpenAI. (2025). ChatGPT (Nov 2025 version) [Large language model]. OpenAI.

<https://chat.openai.com>